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AegisDrive: Adversarially-Constrained End-to-End Driving Model Enhancement Framework

**Datenzentrierte Analyse und Augmentierung zur Verbesserung von
End-to-End-Fahrssystemen unter Out-of-Distribution-Szenarien**

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This work explores the task of improving the robustness of end-to-end autonomous driving models in adversarial collision scenarios. Through targeted dataset design, model adaptation, and safety-oriented evaluation, the study investigates how such models can better handle critical situations without degrading benchmark performance.

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1 Introduction

End-to-end autonomous driving (E2E-AD) systems have shown great promise by directly mapping sensor inputs to driving actions. However, their performance often degrades in rare or out-of-distribution (OOD) scenarios such as near-collisions or unexpected road behaviors—conditions that are underrepresented in state-of-the-art training datasets such as nuScenes [1] and NAVSIM [2]. This thesis investigates the role of data augmentation in addressing this limitation. We aim to improve the robustness and generalization capabilities of end-to-end autonomous driving models by (a) analyzing how OOD scenarios impact the models’ decision patterns, and (b) introducing targeted data augmentations that simulate critical edge cases.

1.1 Motivation

While existing (E2E-AD) systems such as UniAD [3] have achieved strong multi-task performance in standard benchmark datasets like nuScenes [1], demonstrating accurate 3D object detection, effective motion prediction, and reliable closed-loop planning in structured urban environments, they often fail in rarely occurring OOD scenarios such as near-collisions, sudden pedestrian appearances, or rare traffic maneuvers. This vulnerability highlights a critical safety concern: autonomous vehicles must remain reliable not only in common cases but also in unexpected, high-risk situations.

1.2 Problem Statement

There is a lack of systematic evaluation and diagnostic analysis to understand how and why (E2E-AD) systems underperform in conditions not represented during training. This research aims to fill that gap by empirically evaluating the performance of existing models on OOD scenarios they were not trained on and analyzing the underlying causes of failure, whether architectural, data-related, or representational. This diagnostic insight will inform our development of targeted data augmentation strategies to improve model resilience and safety under rare but critical driving conditions.

To address the limitations of existing end-to-end autonomous driving models in handling adversarial collision scenarios, we propose a structured pipeline that combines dataset generation, model adaptation, and safety evaluation. The pipeline begins with the generation of adversarial driving scenarios inspired by the nuScenes and Challengers datasets, where an additional adversarial vehicle is introduced to create potential collision situations. These generated scenes are then integrated with the original dataset to form a mixed training dataset. In the next stage, several state-of-the-art autonomous driving models are finetuned using this

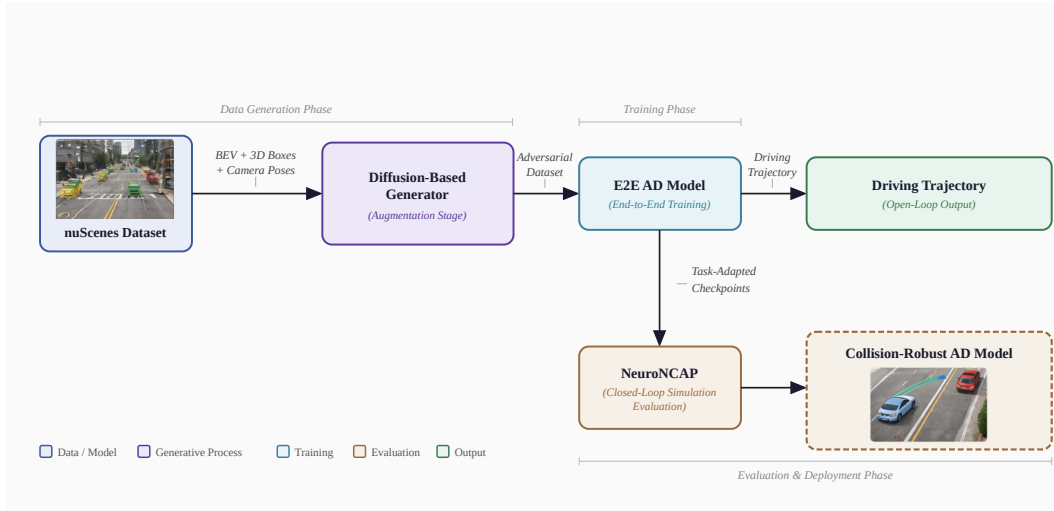


Figure 1.1: Overview of the proposed pipeline for training a collision-robust end-to-end autonomous driving model. The pipeline consists of three phases. In the Data Generation Phase, the nuScenes dataset is fed into a diffusion-based generator to produce an adversarial dataset. In the Training Phase, the adversarial dataset is used to train an end-to-end autonomous driving (E2E AD) model. In the Evaluation and Deployment Phase, task-adapted checkpoints of the trained model are evaluated under closed-loop simulation using NeuroNCAP, resulting in a collision-robust AD model.

dataset to improve their ability to react to collision-prone situations. Finally, the adapted models are evaluated using the NeuroNCAP framework to assess their collision avoidance performance and overall safety behavior. The overall workflow of the proposed approach is illustrated in Figure 1.1.

2 Literature Review

This chapter presents the literature review relevant to this work. It begins with a background section that introduces the key concepts and terminology frequently used throughout the thesis, providing the necessary foundation for understanding the methodologies and models explored in this research. This is followed by a related work section, where existing approaches in end-to-end autonomous driving and adversarial scenario evaluation are reviewed. The goal of this section is to assess the current progress in the field, highlight the limitations of existing methods, and motivate the need for the framework proposed in this thesis.

2.1 Background

2.1.1 Autonomous Driving System Paradigms

Autonomous driving systems have traditionally been designed following a modular pipeline, in which the overall driving task is decomposed into sequential components: perception, tracking, prediction, planning, and control [4] [5] [6] [7]. This architecture reflects classical robotics system design, where each module solves a well-defined subproblem and passes its output to the next stage.

- *Perception*: The perception module transforms raw sensor data (e.g., camera images, LiDAR point clouds, radar signals) into a structured representation of the environment. Perception tasks include:
 - 3D object detection
 - Semantic segmentation
 - Lane and map element recognition
 - Free-space estimation

Modern perception systems rely heavily on deep neural networks, including convolutional and transformer-based architectures [8, 9]. The output of perception can be expressed as:

$$O_t = \{o_1, o_2, \dots, o_n\} \quad (2.1)$$

where each object o_i contains position, orientation, velocity, and semantic attributes.

- *Tracking*: The tracking module associates detected objects across time to maintain consistent identities and estimate dynamic states. This is commonly formulated as multi-object tracking (MOT).

Classical tracking approaches rely on Bayesian filtering methods such as Kalman filters and probabilistic data association [10].

- *Prediction*: The prediction module forecasts future trajectories of surrounding agents over a time horizon T . This task is inherently multimodal and interaction-aware.

$$p(\tau_{t+1:t+T} \mid \mathcal{O}_{1:t}) \quad (2.2)$$

- *Planning*: The planning module computes a feasible and safe future trajectory for the ego vehicle, given the current ego state, perceived environment, predicted trajectories of other agents and the traffic rules and map constraints. The output is a trajectory:

$$\tau_{t+1:t+T}^{ego} \quad (2.3)$$

that satisfies safety and dynamic feasibility constraints.

- *Control*: The control module translates the planned trajectory into executable vehicle commands such as steering, throttle, and braking. Control algorithms commonly include PID controllers, Linear Quadratic Regulators (LQR), and MPC-based controllers [11]. The objective is to minimize trajectory tracking error while respecting vehicle dynamics constraints.

Although modular pipelines provide interpretability and structured development, they introduce error propagation, limited information sharing between tasks, uncertainty accumulation, and high system complexity. These limitations have motivated the development of unified and end-to-end learning-based frameworks.

2.1.2 Scene Representation in Autonomous Driving

Scene representation plays a central role in modern autonomous driving systems, as it determines how spatial, semantic, and dynamic information is structured for downstream tasks such as detection, prediction, and planning.

Among various representations, the bird's-eye-view (BEV) representation has emerged as a dominant design choice in end-to-end autonomous driving frameworks due to its geometric consistency and planning-friendly properties.

Traditional camera-based perception operates in the image plane, where objects are represented in perspective view. While this representation is suitable for 2D detection, it introduces several challenges such as scale distortion due to perspective projection and inconsistent spatial alignment across camera views.

In contrast, BEV provides a top-down, metric-space representation of the environment, where spatial relationships are expressed in a Cartesian coordinate system aligned with the ground plane [12]. This representation offers geometric interpretability since distances and relative positions are directly measurable as well as multi-sensor fusion compatibility as LiDAR data is naturally expressed in BEV [13]. These properties make BEV especially suitable for unified perception–prediction–planning models.

Constructing a BEV representation as shown in Figure 2.2 from camera images requires transforming perspective image features into a spatially aligned ground-plane representation. This process typically consists of two main stages:

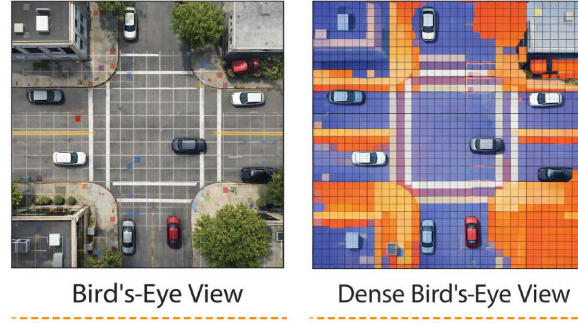


Figure 2.1: Comparison between the standard BEV and dense BEV representations, highlighting the increased spatial density and richer scene representation provided by the dense BEV.

- *Image Feature Extraction*: Each camera image is processed by a convolutional neural network (CNN) backbone (e.g., ResNet or Swin Transformer) to extract high-level semantic features.
- *View Transformation (Image-to-BEV Projection)*: The core challenge lies in lifting 2D image features into 3D space and projecting them onto the ground plane. Several approaches have been proposed such as Depth-Based Lifting [12], this approach uses camera calibration parameters to perform differentiable geometric projection. [9] proposed a transformer-based approach that avoids explicit depth estimation. In BEVFormer, learnable BEV queries are defined in the ground plane. Cross-attention samples image features using projected reference points. Finally, spatial and temporal attention modules aggregate multi-view and multi-frame information. This formulation improves robustness and eliminates explicit voxelization.

Most unified autonomous driving models adopt a *dense bird's-eye-view (BEV) grid representation*, in which the ground plane is discretized into fixed-resolution cells. The difference between the standard BEV representation and the dense BEV representation is illustrated in Figure 2.1. Formally, the BEV feature map can be expressed as

$$\mathbf{F}_{\text{BEV}} \in \mathbb{R}^{H_{\text{bev}} \times W_{\text{bev}} \times C}, \quad (2.4)$$

where H_{bev} and W_{bev} denote the spatial resolution of the grid along the longitudinal and lateral directions, respectively, and C represents the feature dimension. Each grid cell encodes semantic and geometric information corresponding to a small spatial region in the physical world.

This dense BEV paradigm is widely adopted in modern unified driving frameworks, including [3], [14], [15] (as an intermediate representation), and [16].

The dense BEV representation provides uniform spatial reasoning due to its regular grid structure that enables consistent geometric interpretation across the scene. Its tensor formulation is naturally compatible with convolutional neural networks which provides efficient feature extraction and refinement. Moreover, the spatial alignment of the representation allows downstream detection, prediction, and planning modules to operate directly on metric-consistent feature maps.

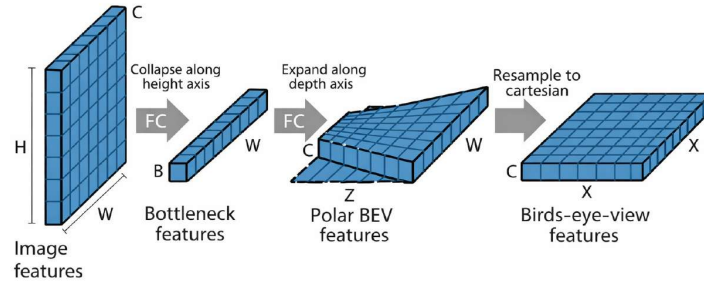


Figure 2.2: Illustration of the transformation from image-space features to bird’s-eye-view (BEV) features. The process begins with convolutional image features represented as a 3D tensor with dimensions height H , width W , and channels C . These features are first collapsed along the height axis using a fully connected (FC) layer to produce a compact bottleneck representation. The bottleneck features are then expanded along the depth axis to generate polar BEV features that encode spatial information with respect to depth Z . Finally, the polar representation is resampled into a Cartesian coordinate system, producing the final BEV feature map with spatial dimensions X and Z , which represents the scene from a top-down perspective. Figure adapted from [17]

However, dense BEV representations also exhibit notable limitations. The computational and memory cost increases quadratically with spatial resolution, making high-resolution grids expensive. In addition, a large portion of the discretized space typically corresponds to empty regions, leading to redundant computation. These limitations have motivated the development of sparse and vectorized scene representations.

2.1.3 Transformer Architectures in Autonomous Driving

Transformer architectures have become a foundational component in modern autonomous driving (AD) systems, particularly in unified end-to-end frameworks. Originally introduced for sequence modeling in natural language processing [18], transformers rely on attention mechanisms to model global dependencies.

Recent autonomous driving models such as BEVFormer [9], UniAD [3], and related unified frameworks adopt transformer-based architectures to jointly model perception, prediction, and planning within a shared latent space.

Self-Attention Mechanism

The core building block of a transformer is the **self-attention** mechanism. Given an input set of tokens $X = \{x_1, x_2, \dots, x_n\}$, self-attention computes pairwise interactions between all tokens. Each token is linearly projected into query (Q), key (K), and value (V) representations:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (2.5)$$

The attention weights are computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V \quad (2.6)$$

where d is the feature dimension.

In autonomous driving, self-attention enables modeling interactions among multiple agents (vehicle–vehicle interactions), agents and map elements, and spatial regions in BEV representations.

Unlike convolutional networks, which have limited receptive fields, self-attention provides global context modeling, which is crucial for understanding complex traffic scenarios.

Cross-Attention Mechanism

Cross-attention extends self-attention by allowing one set of tokens (queries) to attend to another set (keys and values). Formally, given query tokens Q and context tokens (K, V):

$$\text{CrossAttention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V \quad (2.7)$$

In autonomous driving, cross-attention is widely used for projecting image features into BEV space (e.g., BEVFormer [9]), aggregating map features with agent tokens, and conditioning planning queries on predicted agent states. This mechanism enables flexible information fusion across task-specific representations.

Query-Based Modeling

Modern transformer-based AD systems adopt **query-based modeling**, inspired by DETR [19]. In this formulation, learnable query embeddings are introduced to extract structured outputs from latent features.

Common query types in autonomous driving include:

- **Object queries:** represent potential dynamic agents for detection and tracking.
- **Map queries:** represent lane segments, boundaries, or road topology elements.
- **Planning queries:** represent ego trajectory hypotheses or control tokens.

Each query interacts with global scene features via cross-attention and is refined through transformer layers. This approach allows the model to directly predict structured outputs without relying on hand-crafted post-processing pipelines. Transformer architectures provide a flexible and powerful framework for modeling global spatial dependencies, multi-agent interactions, and cross-modal information fusion in autonomous driving. Through self-attention, cross-attention, and query-based modeling, they enable unified end-to-end learning of perception, prediction, and planning within a shared representation space. As a result, transformers have become a core component in modern planning-oriented autonomous driving frameworks.

2.1.4 Motion Prediction and Interaction Modeling

Motion prediction requires modeling multimodal trajectories, agent–agent interactions, agent–map constraints, and temporal dependencies. Modern approaches leverage graph-based and attention-based architectures to capture these complex relationships for safe downstream planning.

Agent–Agent Interaction

Agent behavior is not independent; it is influenced by nearby agents. For example, vehicles slow down when another vehicle merges or yield when pedestrians cross. Capturing these dependencies requires explicit interaction modeling. Interaction modeling is commonly achieved using social pooling mechanisms [20] or graph neural networks (GNNs) where agents are nodes and edges represent interactions [21]

Agent–Map Interaction

Agent behavior is constrained by static scene structure such as lanes and road topology. Vectorized map representations [21] encode lanes as polylines and incorporate geometric constraints.

Temporal Modeling Across Frames

Motion prediction is inherently temporal. Models must capture motion dynamics over multiple past frames to infer velocity and acceleration patterns. Temporal modeling approaches

can be transformer-based temporal attention such as spatiotemporal attention in BEVFormer [9]

2.1.5 Generative Models for Planning

Recent advances in end-to-end autonomous driving have explored generative modeling techniques for trajectory planning. In contrast to deterministic trajectory regression, generative approaches formulate planning as a probabilistic inference problem to model uncertainty and multimodality.

Regression-Based Trajectory Prediction

Traditional learning-based motion planners formulate trajectory prediction as a deterministic regression problem. Given a scene representation S , the model predicts a single future trajectory:

$$\hat{\tau} = f_{\theta}(S) \quad (2.8)$$

The training objective typically minimizes displacement error:

$$\mathcal{L} = \|\tau - \hat{\tau}\|_2^2 \quad (2.9)$$

This approach is common in early trajectory forecasting works such as Social LSTM [20] and MultiPath [22]. While computationally efficient, regression-based models suffer from the multimodality problem. In real-world driving, future behavior is inherently multimodal. At an intersection, for example, a vehicle may turn left, turn right, or continue straight. Deterministic models trained with L2 loss tend to predict the mean of all plausible outcomes, leading to unrealistic averaged trajectories. Multi-hypothesis methods such as MultiPath [22] attempt to address this by predicting multiple anchor trajectories, but expressiveness remains limited.

Diffusion Models

Diffusion models were introduced in [23] and further developed in score-based generative modeling [24]. They learn a data distribution through iterative noise addition and denoising.

- **Forward Diffusion Process:** The forward diffusion process gradually adds Gaussian noise at each step and after many steps, the data distribution converges. During inference, a trajectory is generated by starting from Gaussian noise and iteratively denoising it.
- **Conditional Diffusion:** For planning, diffusion is conditioned on scene context, typically BEV features, agent states, and map information. Conditioning is implemented via feature concatenation or cross-attention.

Deterministic planners are efficient but struggle with multimodality. Diffusion-based generative planning models address this limitation by learning full trajectory distributions and generating samples via iterative denoising.

2.1.6 Simulation-based Evaluation of Autonomous Driving Models

Real-world testing is costly, time-consuming, and potentially unsafe, simulation environments provide a scalable and controlled platform for assessing driving policies under diverse traffic conditions. Simulation enables systematic testing of rare and safety-critical scenarios that may be difficult to capture in naturalistic datasets. This paradigm has been widely adopted in autonomous driving research and development, with platforms such as CARLA [25].

Beyond safety and scalability, simulation allows controlled different scenarios and reproducibility of experiments. These properties are essential for evaluating decision-making systems, whose performance depends not only on instantaneous prediction accuracy but also on the stability of sequential actions over time. Prior work has emphasized that offline accuracy metrics may not reliably reflect driving performance under interactive conditions due to distribution shift and compounding errors. Within simulation, both open-loop and closed-loop protocols can be defined:

- **Open-loop simulation evaluation** replays scenarios where other agents follow scripted or logged trajectories, scoring the ego model against a fixed reference without allowing the ego to meaningfully alter the scenario evolution.
- **Closed-loop simulation evaluation** executes the ego model's predicted controls/trajectory, allowing the world to evolve in response. This captures feedback effects and error accumulation, and supports safety-relevant metrics such as collisions, off-road rate, traffic-rule violations, progress, and comfort [26][25].

Across protocols, AD evaluation faces persistent challenges: defining metrics that reflect safety and comfort, ensuring coverage of rare long-tail events, and addressing simulation fidelity and sim-to-real mismatch [25].

2.2 Related Work

2.2.1 Autonomous Driving Modular Approach

[4] work formulates driving purely as a reinforcement learning (RL) problem. The model takes raw monocular camera images as input and outputs low-level control commands, specifically steering angles, thereby learning the entire driving policy from visual observations alone. Architecturally, the system consists of a convolutional neural network (CNN) that extracts spatial features from the input images, followed by fully connected layers that map these features to continuous control outputs. Unlike supervised learning approaches that depend on large labeled datasets, the model is optimized using a reward-driven objective. The vehicle interacts with the real environment, and the policy is updated to maximize cumulative reward, which is designed to encourage staying within lane boundaries and maintaining smooth driving behavior. To address the data inefficiency and safety challenges of real-world RL, the authors introduce several stabilizing strategies, including reward shaping, safety constraints, and data augmentation techniques to improve generalization. The key contribution of this work lies in demonstrating that a compact neural network can learn lane-following behavior in under a day of on-vehicle training.

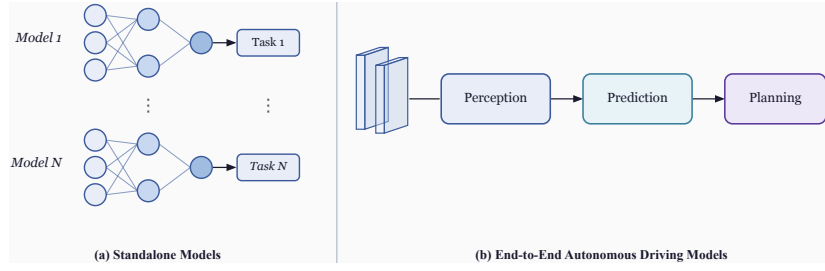


Figure 2.3: Comparison on the designs of autonomous driving framework. the original modular standalone approach vs the end to end approach. Figure adapted from [3].

Another approach that isolates the motion planning component and treats it as a supervised learning problem conditioned on upstream scene understanding is [5]. The authors train a supervised learning-based planner that predicts future ego trajectories from structured scene representations and examine how performance varies under controlled modifications of the training dataset. Their analysis demonstrates that planners trained primarily on common driving scenarios often fail to generalize to rare but safety-critical events.

Although these modular paradigm enables structured system design, it inherently restricts the flow of information across tasks. In particular, valuable contextual and semantic information available in the original sensor data cannot be directly accessed by downstream modules such as planning. Instead, planning relies exclusively on the processed outputs of preceding perception and prediction components. Consequently, errors introduced in earlier stages may propagate through the pipeline, significantly degrading overall system performance. Moreover, such errors cannot be effectively corrected at later stages, which may ultimately compromise driving safety.

2.2.2 End-to-end Autonomous Driving

To address these limitations, recent research has increasingly focused on holistic formulations of the driving task [14] [27] [28], commonly referred to as end-to-end (E2E) autonomous driving. In contrast to modular architectures, E2E models directly map raw sensor inputs, such as camera images, LiDAR point clouds, or radar measurements—to driving actions including steering, acceleration, and braking. Rather than optimizing each task independently, as shown in Figure 2.3 these approaches employ a unified model that jointly learns perception, prediction, planning, and control within a single framework. Typically implemented using deep neural networks, end-to-end systems learn rich spatial and temporal representations from large-scale driving data. By eliminating handcrafted intermediate representations and enabling joint optimization across tasks, E2E approaches aim to mitigate error accumulation, enhance information sharing, and improve robustness in diverse and complex driving scenarios.

One of the first E2E autonomous models was [3]. UniAD proposes a unified, planning-oriented framework that reformulates autonomous driving as a multi-task learning problem centered around motion planning. Instead of treating perception, prediction, and planning as loosely connected modules, UniAD integrates them into a single end-to-end trainable architecture built upon a shared bird’s-eye-view (BEV) representation. As shown in Figure 2.4, the model first extracts multi-view image features using a convolutional backbone and

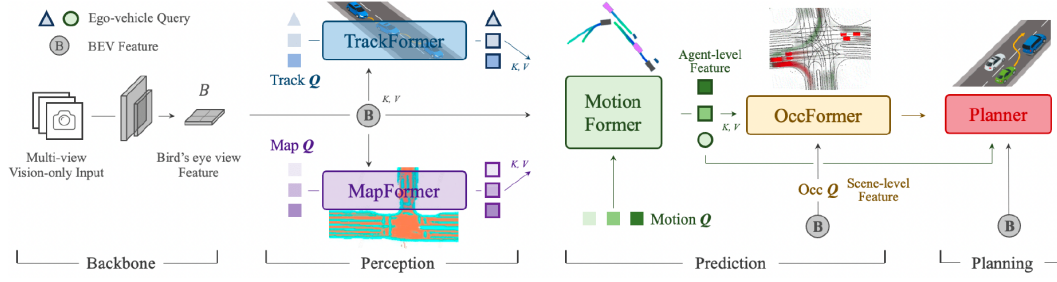


Figure 2.4: Pipeline of Unified Autonomous Driving (UniAD). All perception and prediction modules are designed in a transformer decoder structure, with task queries as interfaces connecting each node. A simple attention-based planner is in the end to predict future waypoints of the ego-vehicle [3]

lifts them into a BEV space via a view transformation module, to enable spatially consistent scene encoding. On top of this BEV feature map, UniAD performs joint detection and tracking using object queries that are temporally propagated across frames, thereby maintaining instance-level consistency. These tracked agents are then fed into a motion forecasting module that models multimodal future trajectories through interaction-aware attention mechanisms. Importantly, the planning module is tightly coupled with upstream tasks: it reasons over the predicted motions of surrounding agents and learns to generate the ego vehicle's future trajectory directly within the same unified feature space. The architecture employs task-specific heads while sharing intermediate representations, and it is trained in a stage-wise yet end-to-end manner with coordinated multi-task losses. By explicitly designing the entire pipeline around planning objectives, UniAD ensures that perception and prediction features are optimized not only for intermediate accuracy but also for their contribution to safe and effective trajectory planning, thereby reducing task inconsistency and mitigating error propagation common in traditional modular systems.

Building on UniAD's planning-oriented unified approach, [29] proposes a unified end-to-end autonomous driving framework that models the environment using structured vectorized tokens instead of dense bird's-eye-view (BEV) feature grids. Instead of discretizing space into rasterized BEV grids, VAD represents the driving scene using structured vectorized elements as shown in Figure :

- Lane boundaries are modeled as polylines parameterized by ordered control points in the ground-plane coordinate system.
- Road topology is represented as connected graph elements that encode structural relationships between lane segments, intersections, and connectivity constraints.
- Dynamic agents are modeled as object-level tokens enriched with geometric and kinematic attributes, such as position, heading, velocity, and bounding box dimensions.

These vector elements are extracted from multi-view camera inputs through a backbone network and subsequently refined via transformer-based attention mechanisms. Each scene component is encoded as a learnable token containing geometric and semantic attributes to enable structured reasoning at the object and map level.

A central transformer module facilitates interaction among vectorized map elements, detected agents, and ego-related queries through self- and cross-attention operations, allowing

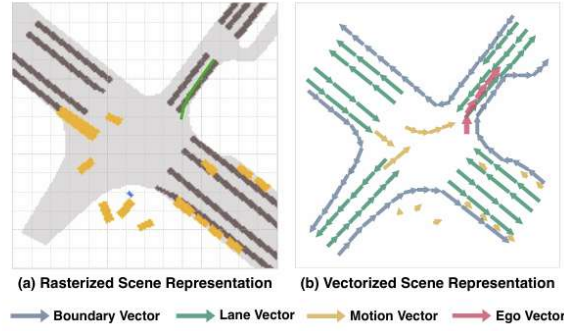


Figure 2.5: Previous paradigms rasterized representation (a) for planning vs the proposed VAD vectorized scene representation (b) for end-to-end planning. Figure adapted from [29]

the model to capture agent-agent and agent-map dependencies efficiently. Detection, motion prediction, and planning are implemented as task-specific heads operating on the shared vector tokens. In particular, future trajectories of surrounding agents are predicted in a multimodal manner, while the ego planner generates future waypoints conditioned directly on the structured scene representation. By operating in a compact vector space, the model significantly reduces computational overhead compared to dense BEV-based approaches while preserving long-range relational reasoning.

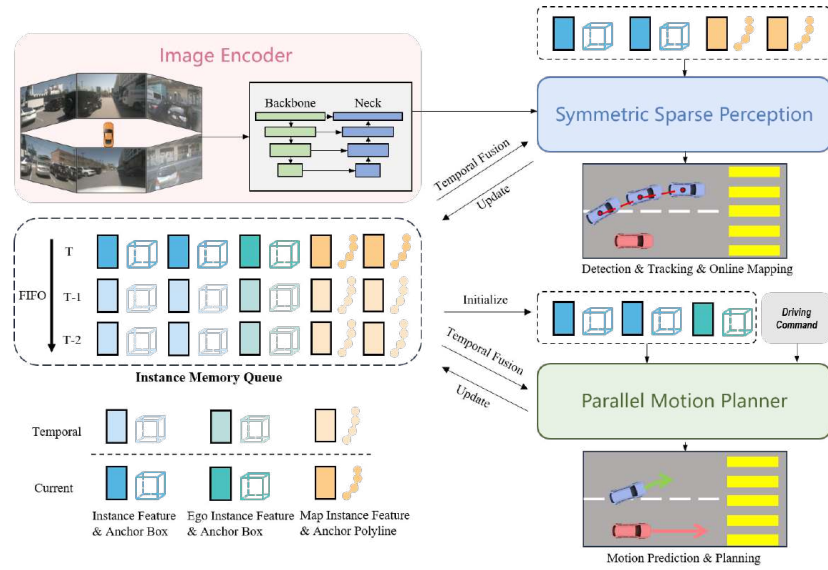


Figure 2.6: Overview of SparseDrive. SparseDrive first encodes multi-view images into feature maps, then learns sparse scene representation through symmetric sparse perception, and finally perform motion prediction and planning in a parallel manner. An instance memory queue is devised for temporal modeling [15]

In a similar effort to improve efficiency and structural reasoning in unified frameworks, SparseDrive [15] proposes an E2E autonomous driving framework that replaces dense bird’s-eye-view (BEV) feature maps with a structured sparse scene representation composed of learnable queries corresponding to key scene entities. Instead of constructing high-resolution dense spatial grids, the model represents the environment through a limited set of object-level and map-level tokens that encode dynamic agents, static map elements, and ego-relevant context. As shown in Figure 2.6 firstly, a convolutional backbone extracts multi-view camera features and lifts them into BEV space. After that, the model employs transformer-based cross-attention to aggregate relevant information into sparse scene queries rather than maintaining dense feature tensors. These queries interact through self-attention layers, enabling explicit modeling of agent-agent and agent-map relationships while significantly reducing computational redundancy. On top of this sparse representation, SparseDrive jointly performs perception, motion prediction, and trajectory planning within a unified architecture. Detection and tracking are formulated through object queries, future motion is predicted via interaction-aware trajectory heads, and the ego planning module generates future waypoints conditioned directly on the shared sparse scene tokens. By aligning all tasks within the same compact representation space, the model facilitates efficient cross-task information exchange and mitigates error accumulation typical of modular pipelines.

Moving beyond deterministic trajectory regression, DiffusionDrive [16] reformulates end-to-end autonomous driving as a conditional generative modeling problem, where the ego vehicle’s future trajectory is sampled from a learned diffusion process conditioned on scene representations. Instead of directly regressing a single deterministic trajectory, the model treats planning as an iterative denoising process that progressively refines a noisy trajectory into a feasible and context-aware future path. As shown in Figure 2.7 The framework first extracts scene features from multi-view camera inputs using a backbone network and projects

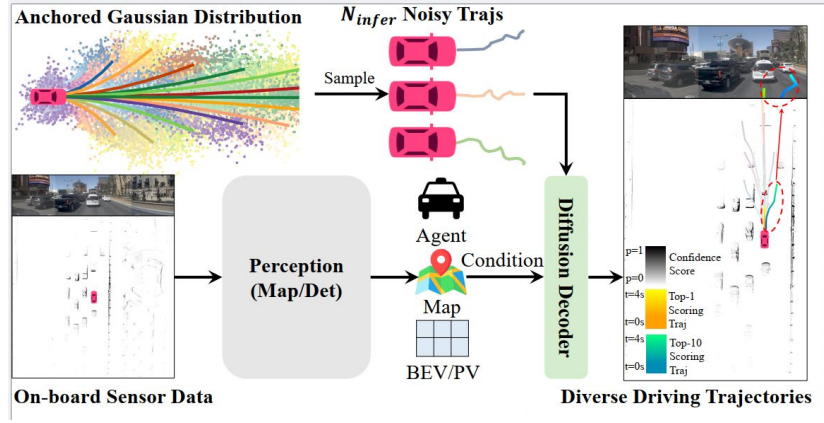


Figure 2.7: Overall architecture of DiffusionDrive. DiffusionDrive integrates various existing perception modules and sensor inputs, The designed diffusion decoder takes the sampled noisy trajectories from anchored Gaussian distribution as input and progressively denoises them with enhanced interactions with the conditional scene context in a cascade manner to generate the final predictions. [16]

them into a bird’s-eye-view (BEV) representation. On top of this shared representation, perception and motion prediction modules encode dynamic agents and map context, providing structured conditioning signals for the planner similar to SparseDrive.

The core contribution lies in the truncated diffusion planning head. During training, ground-truth future trajectories are gradually perturbed with Gaussian noise over multiple diffusion steps, and the model learns to predict the noise residual at each step conditioned on the scene context. During inference, rather than running the full diffusion chain—which would be computationally expensive for real-time driving, the model performs a truncated reverse diffusion process with a reduced number of denoising steps. This significantly improves efficiency while retaining the ability to model multimodal trajectory distributions. By generating trajectories through iterative refinement, DiffusionDrive reduces mode collapse typically observed in deterministic planners. Furthermore, the diffusion formulation naturally integrates uncertainty modeling into the planning process, enhancing robustness in ambiguous or interactive traffic scenarios.

Another generative planning approach is [30] which focuses on modeling historical context to improve both motion prediction and ego trajectory planning. Unlike approaches that primarily rely on the current frame or short temporal windows, this model incorporates extended historical information of both ego and surrounding agents to enhance scene understanding and behavioral forecasting. Multi-view camera inputs are first encoded in a bird’s-eye-view (BEV) representation, after which temporal features from previous frames are aggregated using transformer-based attention mechanisms. This temporal aggregation preserves motion continuity over time, forming a richer scene representation. Instead of predicting only future trajectories, the model is trained to reconstruct or forecast intermediate historical states, effectively enforcing temporal consistency in the learned latent space. This bidirectional temporal constraint strengthens the coupling between perception, prediction, and planning by ensuring that learned representations capture coherent motion dynamics. The motion prediction module generates multimodal future trajectories for surrounding agents based on temporally enriched features, while the planning head produces the ego vehicle’s future trajectory condi-

tioned on both historical motion patterns and predicted agent behaviors. By explicitly bridging past dynamics with future outcomes, the model mitigates abrupt or inconsistent trajectory generation and improves robustness in interactive driving scenarios.

While BridgeAD emphasizes temporal consistency between past and future, Divide and Merge [31] addresses a different limitation of unified models by explicitly disentangling motion dynamics and semantic scene understanding. The paper decomposes scene representation into two complementary branches, motion learning and semantic learning—before merging them for joint planning. The central motivation is that existing unified models often entangle geometric semantics (e.g., lane structures, object categories) and motion dynamics (e.g., velocity, interaction patterns) within a single latent space, which may hinder specialized reasoning. To address this, the framework first extracts multi-view image features and projects them into (BEV) representation. This shared backbone then feeds into two parallel streams: a semantic branch that focuses on static and structural scene elements such as lanes and object identities, and a motion branch that captures temporal evolution, agent kinematics, and interaction cues across frames.

The motion branch incorporates temporal modeling modules, typically transformer-based or recurrent attention mechanisms, to encode dynamic relationships among agents over time. In contrast, the semantic branch emphasizes spatial reasoning and scene layout through attention over static map features and object-level embeddings. After independent feature refinement, a merge module integrates both representations via cross-attention, enabling the planner to reason about what entities exist in the scene (semantic understanding) and how they are moving (motion understanding). The unified planning head then generates the ego vehicle's future trajectory conditioned on this fused representation. By explicitly separating and subsequently integrating motion and semantic information, the model improves representation disentanglement and enhances interaction modeling.

2.2.3 Vision Language Models in Autonomous Driving

Recent advances in large-scale vision-language models (VLMs) have inspired a new line of research in autonomous driving that integrates visual perception with natural language understanding and generation. Unlike traditional autonomous driving systems that operate purely on structured representations such as bounding boxes, trajectories, or BEV features, vision-language-based approaches aim to enable semantic reasoning, explainability, and instruction-following capabilities. These models typically leverage pretrained multimodal foundation models and adapt them to driving tasks such as decision making, scene understanding, and trajectory generation.

DriveGPT4 [32] extends large vision-language foundation models to autonomous driving by enabling conversational scene understanding and high-level decision reasoning. The model integrates visual inputs such as camera images or BEV representations with a large language model backbone. It is designed to generate natural language descriptions of driving scenes, explanations of driving decisions, and high-level driving intentions. It treats autonomous driving as a multimodal reasoning problem and enables interactive querying of the driving system. This makes it particularly suitable for explainable AI applications and human-in-the-loop systems. However, DriveGPT4 typically does not operate as a fully end-to-end real-time

planning system, and its outputs often require downstream modules for trajectory generation and control execution.

ImpromptoVLA [33] focuses on instruction-conditioned driving and aims to bridge perception, language understanding, and control generation within a unified architecture. Unlike DriveGPT4, which primarily generates reasoning outputs, ImpromptoVLA is trained to map visual inputs and textual prompts directly to driving actions. The model leverages pretrained vision-language encoders and fine-tunes them for driving-specific tasks. Language prompts include commands such as turning left at the next intersection, yielding to pedestrians, or parking beside a specified vehicle. By conditioning action generation on language inputs, ImpromptoVLA enables flexible goal specification and interactive planning. However, such models depend heavily on the diversity and coverage of language-conditioned training data, and generalization to unseen instructions or rare traffic scenarios remains challenging.

OpenDriveVLA [34] extends the vision-language-action paradigm with an emphasis on open-world generalization. It integrates large-scale pretrained multimodal models with driving-specific fine-tuning to handle diverse environments. The framework learns a unified visual-linguistic embedding space and predicts actions conditioned on both scene context and language instructions. Compared to ImpromptoVLA, OpenDriveVLA places stronger emphasis on robustness and scalability through large-scale pretraining. Nevertheless, performance remains sensitive to domain shift between internet-scale multimodal data and structured driving environments.

Although all three models leverage multimodal representations, their primary distinctions lie in their objectives and integration levels. DriveGPT4 emphasize reasoning and explainability, whereas ImpromptoVLA and OpenDriveVLA aim to directly integrate language conditioning with action generation. Despite their promising capabilities, vision-language models face several fundamental limitations in the autonomous driving paradigm. Large language backbones introduce computational overhead that may conflict with real-time constraints. Language-based reasoning outputs are difficult to formally verify, making safety certification challenging. These models may also produce semantically plausible but physically infeasible decisions if not tightly constrained by geometric planning modules. Furthermore, most vision-language models are pretrained on internet-scale datasets that differ significantly from structured traffic environments, leading to domain adaptation challenges. Finally, language-driven architectures typically operate at a semantic decision level and may lack the precision required for continuous trajectory optimization.

2.2.4 Dataset Generation Models

Recent progress in generative modeling has made driving video synthesis increasingly relevant for autonomous driving research, especially for dataset augmentation and safety-oriented scenario generation. In contrast to conventional data collection, video generation methods can create rare, risky, or controllable situations at substantially lower cost, making them particularly attractive for improving the robustness of end-to-end autonomous driving models.

Challenger Challenger [35] is designed specifically for generating *adversarial* driving videos that are both photorealistic and physically plausible. Its central motivation is that many prior adversarial-driving approaches operate only in abstract representations such as trajectories or BEV layouts, which are insufficient for fully stress-testing camera-based end-to-end autonomous driving systems. To address this, Challenger separates the problem into two coupled stages: adversarial trajectory construction and photorealistic video synthesis. The method first searches for candidate adversarial maneuvers using a physics-aware multi-round trajectory refinement procedure, then scores these trajectories using a tailored objective that balances adversarial difficulty, realism, and compatibility with downstream rendering. After identifying suitable adversarial traffic interactions, the resulting trajectories are rendered into multi-view driving videos using a high-fidelity video generation backbone. The paper demonstrates scenarios such as cut-ins, sudden lane changes, tailgating, and blind-spot intrusions, and shows that the generated videos significantly increase the collision rate of several end-to-end autonomous driving models. :contentReference[oaicite:1]index=1

Technically, the key contribution of Challenger is not merely video synthesis quality, but the explicit coupling between *adversarial scenario search* and *sensor-realistic rendering*. This makes it especially relevant for safety evaluation: instead of generating generic diverse traffic videos, Challenger deliberately constructs traffic interactions that are likely to induce planning failures. In that sense, it is closer to a targeted scenario-generation framework than to a purely generative world model.

Delphi Delphi [36] addresses a different but closely related problem: how to generate *long*, *controllable*, and *temporally consistent* multi-view driving videos that are useful for improving end-to-end autonomous driving, especially the planning component. The authors argue that earlier generative methods often produce clips shorter than eight frames and suffer from spatial or temporal inconsistencies, which limits their usefulness for planning-oriented models that require stable scene evolution over longer horizons. To overcome this, Delphi proposes a diffusion-based long video generation framework with two notable components: a shared noise modeling mechanism across multiple camera views to improve spatial consistency, and a feature-aligned module that improves controllability and temporal coherence. According to the paper, Delphi can generate sequences up to 40 frames long while preserving consistency, which is substantially longer than prior methods emphasized in the work.

A second important idea in Delphi is that video generation is not used uniformly across the dataset. Instead, the paper proposes a failure-case-driven sampling policy, assisted by pre-trained vision-language models, to identify valuable scenarios for augmentation. In other words, Delphi is not only a video generator but also a data-selection framework: it tries to

generate new samples that resemble model failure cases so that the synthetic data has maximal training utility. This is particularly relevant for end-to-end driving because the goal is not just to increase visual diversity, but to create long-horizon samples that improve planning robustness. The paper reports that generating synthetic data amounting to only a small fraction of the original training set can produce meaningful gains, including improvements in planning performance. Relative to Challenger, Delphi is less explicitly adversarial and more oriented toward *generalization through targeted augmentation*; relative to DriveDreamer-2, it is less language-centric and more focused on long-horizon consistency and failure-driven sample efficiency.

DriveDreamer-2 DriveDreamer-2 [37] approaches the problem from the perspective of controllable world modeling. Rather than beginning with adversarial optimization or failure-case mining, it introduces an LLM-enhanced pipeline that allows a user to specify a desired driving scenario in natural language. The method first uses an LLM interface to convert the user query into agent trajectories. It then generates an HD map that is consistent with those trajectories and traffic rules, and finally synthesizes multi-view videos using a Unified Multi-View Model designed to improve temporal and spatial coherence. This decomposition is technically important because it turns free-form textual intent into a structured driving scene generation process: text is translated into motion, motion into scene layout, and scene layout into video. The main strength of DriveDreamer-2 is therefore *user-level controllability*. It is particularly effective for generating customized or uncommon driving events, including rare maneuvers such as abrupt cut-ins, in a way that is easier to specify than manually engineering trajectories or searching adversarially. The paper also reports improvements in downstream perception tasks such as 3D detection and tracking, alongside improved video-generation quality metrics. However, compared with Delphi, the emphasis is less on long-horizon planning-specific augmentation, and compared with Challenger, the emphasis is less on explicitly maximizing adversarial pressure against autonomous driving models. For this reason, DriveDreamer-2 can be viewed as the most flexible scenario authoring framework among the three, whereas Challenger is the most safety-targeted and Delphi is the most directly connected to improving end-to-end planning robustness through long-video augmentation.

2.2.5 AD testing and evaluation models

Beyond traditional open-loop and closed-loop evaluation protocols, recent research has proposed dedicated evaluation frameworks designed to bridge the gap between offline dataset benchmarking and full interactive simulation. These frameworks aim to provide scalable, reproducible, and safety-aware testing environments while avoiding the computational overhead and engineering complexity of high-fidelity simulators. One representative example is NavSim, introduced in [2], which proposes a hybrid evaluation paradigm that approximates closed-loop interaction using logged data. The core idea is to combine logged real-world driving data with a learned or structured world model that approximates how the environment would evolve under alternative ego actions.

Traditional offline evaluation suffers from a fundamental limitation: the dataset only contains trajectories resulting from expert behavior. If a planning model produces a trajectory

that deviates from the logged expert path, the dataset does not provide ground-truth sensor observations or agent reactions for that counterfactual state. Unlike full simulators such as CARLA [25], NavSim does not model physics or reactive behaviors for non-ego agents. Instead, it reuses recorded trajectories, significantly reducing computational cost and engineering complexity.

NavSim evaluates planning performance using metrics such as collision rate, off-road rate and traffic rule violations. The framework occupies an intermediate position between open-loop evaluation and full simulation. It captures ego-centric closed-loop behavior while approximating environment dynamics via log replay. However, since other agents do not react to ego deviations, interaction modeling remains partially constrained.

While pseudo-simulation frameworks provide scalable approximations of closed-loop behavior, recent work has emphasized the need for high-fidelity, photorealistic evaluation environments that reflect real-world perception challenges. In particular, safety validation of autonomous driving systems requires not only behavioral correctness but also robustness under realistic sensor inputs. To address this need, NeuroNCAP [26] introduces a neural rendering-based evaluation framework that combines closed-loop simulation with photorealistic scene generation. Using neural rendering techniques, the simulator generates images that more closely match real sensor distributions. The evaluation loop proceeds as follows at each time step, the model receives photorealistic rendered camera inputs generated from the neural scene representation. Based on these observations, it predicts control commands or future trajectories for the ego vehicle. The simulator then updates the ego vehicle's state accordingly, modifying its pose and position within the scene. Given this updated view-point, the neural rendering module synthesizes a new camera observation consistent with the ego vehicle's motion. This perception–action cycle continues iteratively, enabling interactive rollouts under realistic visual conditions. Safety and performance metrics are subsequently computed from the resulting trajectory and environmental interactions. Inspired by standardized vehicle safety assessment programs, the framework focuses on safety-critical scenarios such as emergency braking, cut-in maneuvers, pedestrian crossings, and obstacle avoidance. Performance is measured using safety-oriented metrics including collision rate, time-to-collision, minimum distance to obstacles, and traffic rule violations. By combining photorealistic rendering with interactive rollouts, NeuroNCAP aims to reduce the sim-to-real gap and provide a realistic benchmark for safety validation of autonomous driving systems. Although neural rendering introduces computational overhead, it represents a significant step toward high-fidelity closed-loop safety testing.

3 Methodology

While recent autonomous driving architectures demonstrate strong performance on standard benchmarks, their behavior in safety-critical situations such as imminent collisions remains less explored. To address this limitation, the proposed methodology focuses on augmenting existing models with targeted training data that emphasizes such high-risk situations.

The methodology is structured around two main components. First, a custom dataset is designed based on the structure of the nuScenes [1] and Challengers datasets [35], where additional adversarial scenarios are introduced to expose models to high-risk interactions they would otherwise rarely encounter during training. Second, several state-of-the-art end-to-end autonomous driving models, including DiffusionDrive [16], SparseDrive [15], and UniAD [3] are finetuned using this dataset to adapt their learned representations toward safer and more robust behaviour in such conditions. Following this augmentation and finetuning stage, the models are evaluated within the NeuroNCAP evaluation framework to assess their safety performance using both trajectory prediction and collision-based metrics.

3.1 Adversarial Dataset Generation

The dataset generation pipeline is designed to augment the nuScenes dataset with safety-critical adversarial scenarios that are absent from standard autonomous driving benchmarks. The process is inspired by the Challenger [35] and MagicDrive [38] frameworks, and follows a structured pipeline that moves from geometric scene construction through adversarial agent injection to photorealistic multi-camera rendering.

3.1.1 Scene Representation and Base Data

Each scene in the pipeline is represented in two complementary forms: a Bird’s-Eye View (BEV) road map and a set of 3D bounding boxes. The BEV road map encodes the structural layout of the driving environment, including lane boundaries, drivable surface regions, road topology, and static infrastructure elements. The 3D bounding boxes provide the spatial configuration of all dynamic agents present in the scene, encoding their position, dimensions, and orientation in the world coordinate frame. Both representations are sourced directly from the nuScenes dataset annotations, which provides a rich and diverse set of real-world urban driving scenes across a variety of conditions.

This structured geometric representation serves as the foundational scaffold for all downstream generation steps. Unlike approaches that rely solely on projected 2D representations, the use of 3D bounding boxes is essential for preserving accurate object geometry across camera viewpoints, as BEV-only conditioning introduces ambiguity in the vertical axis that

degrades the fidelity of occlusion patterns, object dimensions, and road surface elevations in the rendered output [38].

3.1.2 Adversarial Agent Injection

Once the base scene representation is established, an adversarial vehicle is introduced into the scene by modifying the 3D bounding box set and the corresponding BEV layout. Two distinct adversarial scenarios are considered, each designed to expose a different failure mode in end-to-end autonomous driving models:

Frontal Collision Scenario.

In this scenario, an adversarial vehicle is placed in the ego vehicle's lane and assigned a trajectory that approaches the ego vehicle head-on from the front. The adversarial agent's 3D bounding box is initialised at a distance ahead of the ego vehicle and its trajectory is defined as a sequence of poses moving progressively closer along the longitudinal axis of the ego lane. This simulates a direct oncoming threat that requires the autonomous system to detect an approaching vehicle and respond with an appropriate evasive or braking manoeuvre. The trajectory is constructed to be physically plausible, maintaining consistent heading and speed profiles that reflect realistic vehicle dynamics, following the physics-aware trajectory design principles of the Challenger framework [35].

Stationary Obstacle Scenario.

In this scenario, an adversarial vehicle is placed at a close distance directly ahead of the ego vehicle and assigned a stationary pose throughout the scene. The vehicle's 3D bounding box is fixed at a position within the ego lane at a proximity that would require an immediate braking or avoidance response. This scenario tests the model's ability to detect and respond to unexpected stationary obstacles that would not be present in routine driving distributions, representing the class of hazards most likely to be underrepresented in naturalistic datasets such as nuScenes.

In both cases, the adversarial vehicle is introduced purely at the annotation level as an additional 3D bounding box with the corresponding trajectory metadata prior to any rendering step. This ensures that the geometric conditioning passed to the rendering stage is consistent with the intended adversarial interaction.

3.1.3 Photorealistic Scene Rendering

With the adversarial annotations in place, the modified scene representation comprising the BEV road map, the updated set of 3D bounding boxes including the adversarial agent, and the camera pose parameters are all passed to a geometry-conditioned diffusion model for photorealistic rendering, following the approach of MagicDrive [38]. The diffusion model synthesises multi-camera street-view images that are consistent with the provided geometric

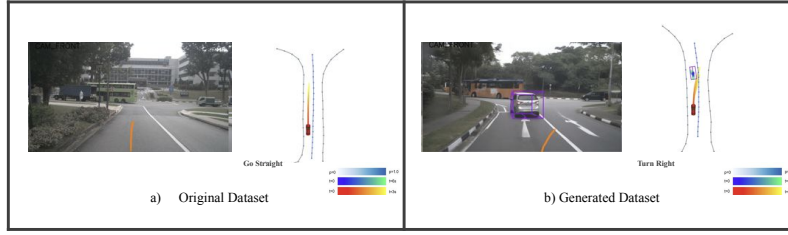


Figure 3.1: Comparison between an original scene from the nuScenes dataset and the corresponding adversarially generated scenario. (a) Original dataset: the unmodified scene with its RGB image and bird’s-eye view representation. (b) Generated dataset: the same scene after inserting an adversarial vehicle to create a collision-prone interaction with the ego vehicle.

conditions, producing realistic visual observations from the perspective of the ego vehicle’s sensor suite.

The use of a diffusion-based renderer is critical for two reasons. First, it ensures that the injected adversarial vehicle appears with realistic appearance, lighting consistency, and correct occlusion behaviour relative to the scene geometry, rather than as a crudely composited object. Second, the cross-view attention mechanism employed in the rendering architecture enforces consistency across the multiple cameras of the nuScenes sensor rig simultaneously, which is a hard requirement for end-to-end autonomous driving models that consume surround-view camera inputs.

3.1.4 Output Format and Compatibility

The output of the pipeline is a set of augmented scenes, each consisting of synchronised multi-camera image sequences, updated 3D annotations, and BEV map data, formatted to be compatible with the standard nuScenes data schema. This ensures that the generated adversarial scenes can be consumed directly by the target E2E models without requiring any architectural modification to their data loaders. The final dataset is composed of the original nuScenes training scenes combined with the adversarially augmented scenes, providing a mixed distribution that exposes the models to both nominal and safety-critical driving conditions during fine-tuning. Figure 3.1 illustrates the generated adversarial scenarios used in our dataset. The left side of the figure shows an example scene taken directly from the original nuScenes dataset while the right side shows the same scene after introducing an adversarial vehicle designed to create a potential collision scenario. In the generated version, the BEV representation additionally illustrates the modified ego trajectory, which reflects how the ego vehicle would respond to the newly introduced obstacle. Instead of continuing along its original straight path, the planned trajectory is adjusted to steer away from the adversarial vehicle, resulting in a collision-avoidance maneuver. Consequently, the corresponding high-level driving command is also updated from *go straight* in the original scene to *turn right* in the generated scenario. This modification preserves the original environmental context while introducing a controlled safety-critical interaction that requires the autonomous driving model to adapt its planning behavior.

3.2 E2E Model Adaptation

Following the construction of the adversarial dataset, the next stage focuses on adapting the selected end-to-end autonomous driving models to these newly introduced safety-critical scenarios. Although the considered models differ in their planning formulations, they all integrate perception, scene understanding, motion reasoning, and trajectory planning within a unified framework. Therefore, rather than retraining each model entirely, the finetuning strategy was designed to selectively adapt the components most directly related to ego trajectory generation under collision-prone conditions.

The main objective of this stage is to preserve the pretrained perception and scene understanding capabilities of the models while improving their planning response to adversarial interactions. In particular, the finetuning process targets the ability of the models to modify their predicted future trajectory and high-level driving command when encountering an inserted adversarial vehicle. The following subsections describe, for each model, the specific adaptation strategy, the components that were frozen or updated during training, and the rationale behind these design choices.

3.2.1 SparseDrive Finetuning Strategy

To evaluate the impact of adversarial scenarios on planning behavior, the SparseDrive model [15] was finetuned using the generated dataset while preserving its perception and scene understanding capabilities. Since the pretrained model already demonstrates strong performance on perception-related tasks such as object detection and motion prediction, retraining these components was not necessary for the objectives of this work.

Instead, the finetuning process focused exclusively on adapting the model’s planning component to the newly introduced adversarial scenarios. Specifically, the image backbone, feature pyramid network (FPN), depth estimation branch, detection head, and map head were frozen during training. This ensured that the perception pipeline remained unchanged while allowing the planning module to learn new trajectory adjustments required to avoid the inserted adversarial vehicles.

This freezing strategy was implemented through the optimizer configuration by assigning a learning rate multiplier of zero to all components except the motion planning head. As a result, only the parameters responsible for ego trajectory planning were updated during training. The goal of this design choice was to encourage the model to modify its trajectory prediction behavior without affecting its ability to detect surrounding agents or understand the scene structure.

The finetuning was performed using a combination of the generated adversarial dataset and a balanced subset of the original nuScenes dataset. This mixed training strategy helps maintain the model’s general driving behavior while enabling it to adapt to the newly introduced safety-critical scenarios. During training, the model learns to adjust its predicted ego trajectory and corresponding driving command when encountering adversarial vehicles that create potential collision situations.

By restricting optimization to the planning head while keeping the perception modules fixed, the finetuning process specifically targets improvements in collision avoidance behavior

without degrading the model’s original perception performance.

3.2.2 DiffusionDrive Finetuning Strategy

The finetuning strategy for DiffusionDrive [16] followed the same general principle adopted for SparseDrive, namely preserving the pretrained perception and scene understanding modules while adapting the planning component to the adversarial scenarios introduced in the generated dataset. Since DiffusionDrive shares the same fundamental image backbone and multi-task structure as SparseDrive, the image backbone, image neck, depth branch, detection head, and map head were again frozen during training. This ensured that the model retained its original capabilities in perception, object detection, and map understanding, while the finetuning process focused on improving the ego trajectory planning behavior under collision-prone situations.

However, in addition to freezing the perception-related modules, the interaction layers within the motion planning head were also frozen. In DiffusionDrive, these interaction layers are responsible for aggregating contextual information from the ego vehicle history, surrounding dynamic agents, and scene structure before the diffusion-based trajectory refinement stage. Their role is to construct an intermediate planning representation that captures how the ego vehicle should reason about its environment and nearby actors. In other words, they provide the contextual embedding upon which the diffusion process later generates and refines candidate future trajectories.

Freezing these interaction layers was beneficial for the current setup because the objective was not to relearn general scene interaction patterns, but rather to adapt the final trajectory generation process to the newly introduced adversarial collision scenarios. The pretrained interaction layers already encode meaningful relationships between the ego vehicle, other traffic participants, and the road layout. Updating them on a comparatively small adversarial finetuning dataset could risk overwriting these stable interaction priors and lead to unnecessary degradation of the model’s general planning behavior. By keeping them fixed, the model preserves its pretrained contextual reasoning, while the trainable diffusion-based refinement components can specialize in producing safer collision-avoidance trajectories.

This design therefore separates two functions within the planner: the interaction layers remain responsible for extracting stable scene and agent context, whereas the diffusion-related refinement layers are allowed to adapt to the new planning targets. Such a setup is particularly suitable for this work, where the adversarial dataset modifies the desired ego response rather than the underlying scene semantics. The main requirement is for the model to generate a different future maneuver, such as steering away from an inserted vehicle, without changing how it fundamentally perceives and interprets the environment.

As in the SparseDrive finetuning setup, training was carried out using a mixed dataset composed of the generated adversarial scenes and a balanced subset of the original nuScenes data. This helps maintain generalization to standard driving situations while exposing the planner to the newly introduced safety-critical cases. The model was trained for 30 epochs using the AdamW optimizer with cosine annealing learning rate scheduling. Compared with SparseDrive, a smaller learning rate was used, which is appropriate given the more selective parameter updates and the need to preserve the stability of the pretrained diffusion-based planning process.

Through this selective freezing strategy, DiffusionDrive was finetuned to improve collision-avoidance planning while minimizing changes to the pretrained perception, interaction reasoning, and broader scene understanding components.

3.2.3 UniAD Finetuning Strategy

In addition to SparseDrive and DiffusionDrive, the UniAD model [3] was also included in the experimental study to provide a complementary planning paradigm for end-to-end autonomous driving. Unlike SparseDrive-based approaches, which rely on command-conditioned trajectory generation using explicit high-level driving commands such as *turn right*, *turn left*, or *go straight*, UniAD follows a unified perception–prediction–planning framework that produces ego trajectories directly from scene understanding without relying on predefined command constraints. This difference makes UniAD particularly useful for evaluating how planning behavior adapts to adversarial scenarios when the model is not explicitly guided by discrete driving commands.

The models based on SparseDrive and DiffusionDrive utilize command-conditioned planning, where the predicted trajectory is influenced by high-level navigation commands. While this design provides strong control over the intended maneuver, it can introduce rigid constraints when adapting to unexpected or adversarial situations. For instance, in the adversarial frontal collision scenarios introduced in the generated dataset, the ego vehicle may need to deviate from its initially commanded path in order to safely avoid the inserted vehicle. When a model is strongly conditioned on commands such as *go straight*, it becomes more difficult for the planner to generate trajectories that violate this instruction, potentially limiting its ability to react to the adversarial obstacle.

UniAD addresses this limitation by generating trajectories directly from the learned scene representation without relying on explicit command inputs. Instead, the planning module reasons jointly over detected objects, predicted agent motions, and the surrounding map context to infer a safe future trajectory. As a result, the model has more flexibility to modify its driving behavior when encountering unexpected obstacles or adversarial agents.

A similar finetuning strategy was adopted to ensure a fair comparison across models. The pretrained perception backbone and intermediate task heads were frozen during training, including the detection, map understanding, and motion prediction modules. This design choice ensures that the model retains its pretrained scene understanding capabilities while allowing the planning module to adapt to the adversarial driving scenarios introduced in the dataset. By restricting parameter updates primarily to the planning components, the finetuning process focuses on improving the model’s trajectory generation behavior without altering the underlying perception pipeline.

The training procedure again used a mixed dataset consisting of the generated adversarial scenes and a balanced subset of the original nuScenes data. This helps preserve the general driving behavior learned from the original dataset while enabling the model to adapt to safety-critical interactions involving adversarial vehicles. Through this setup, UniAD provides an alternative planning formulation that allows us to investigate whether removing explicit command constraints leads to improved robustness in collision avoidance scenarios.

4 Evaluation

4.1 Datasets

4.1.1 Nuscenes Dataset

The **nuScenes dataset** NuScenes [1] has become one of the most widely used benchmarks for 3D object detection, multi-object tracking, motion prediction, and end-to-end autonomous driving research. It was introduced to overcome limitations of earlier datasets by providing rich sensor diversity, full 360-degree coverage, dense annotations, and high-definition map information. Due to its scale and comprehensive annotation protocol,

NuScenes contains 1,000 driving scenes recorded in urban environments in Boston (USA) and Singapore. Each scene is approximately 20 seconds long, resulting in about 20 hours of total driving data. The dataset is divided into *700 training scenes*, *150 validation scenes*, and *150 test scenes*. Test annotations are withheld and evaluation is performed through an official benchmark server to ensure fair comparison.

A defining characteristic of nuScenes is its multimodal sensor suite. Each vehicle is equipped with six cameras providing 360-degree coverage, one 32-beam LiDAR sensor, five radar sensors, GPS, and an IMU. The cameras capture images at 12 Hz, while LiDAR operates at 20 Hz. For most perception tasks, keyframes are *sampled at 2 Hz*, with synchronized data across all sensors. This setup enables research in camera-only, LiDAR-only, radar-based, and sensor-fusion methods.

In terms of annotation, nuScenes provides 3D bounding boxes for *23 object classes*, with *10 primary classes* used for the official detection benchmark. Each annotation includes 3D position, size, orientation, velocity, and additional attributes such as visibility level and object activity. The dataset contains approximately 1.4 million annotated 3D bounding boxes. Instance-level tracking IDs are provided across frames, making the dataset suitable for multi-object tracking tasks.

Additionally, nuScenes offers high-definition map expansions for both cities. These maps contain *vectorized semantic layers* such as lane boundaries, drivable areas, pedestrian crossings, and traffic lights. The availability of map information supports research in trajectory prediction and planning.

In this work, the nuScenes dataset was selected as the primary training and evaluation benchmark, as all the widely recognized baseline models considered in this work—SparseDrive, DiffusionDrive, and UniAD—were originally developed and validated on nuScenes. Using nuScenes ensured consistency with prior work and enabled fair comparison against published results. Its large-scale multimodal sensor setup, standardized evaluation metrics made it suitable for holistic autonomous driving models that jointly address perception, prediction, and

planning. Furthermore, nuScenes was not only used for initial model alignment but also as a validation benchmark. After implementing architectural modifications and fine-tuning strategies tailored to the specific objectives of this work, all models were re-evaluated on nuScenes to ensure that performance improvements were not achieved at the expense of benchmark generalization. This step was crucial to verify that the proposed enhancements improved the targeted task while maintaining competitive performance on standardized nuScenes metrics and ensures that the contributions of this work do not overfit to a narrow objective, but instead remain consistent with established autonomous driving evaluation protocols.

4.1.2 Adversarial nuScenes (Challenger Dataset)

The adversarial nuScenes dataset introduced in [35] is a synthetic adversarial extension built on top of the original nuScenes benchmark. Instead of collecting new real-world data, the authors generate photorealistic adversarial driving videos from existing nuScenes scenes. The goal is to evaluate the robustness of autonomous driving models under visually plausible yet adversarially manipulated conditions.

Structurally, the dataset inherits the core organization of nuScenes. It preserves the scene-based structure, temporal continuity, and multi-camera 360-degree setup, and map information remain aligned with the original data, enabling the reuse of standard nuScenes evaluation metrics without modification.

In terms of size, the adversarial dataset mirrors a *subset of the original nuScenes splits* namely the validation set, generating adversarial counterparts for selected scenes rather than recreating the entire dataset. This makes the approach computationally efficient while maintaining sufficient scale for robustness evaluation.

The generation pipeline is based on adversarial video synthesis. Rather than applying independent pixel-level perturbations, the method produces temporally consistent adversarial modifications across entire sequences. A generative model synthesizes realistic perturbations that preserve scene plausibility while inducing performance degradation in downstream perception or planning systems. Spatial and temporal coherence constraints are enforced to avoid unrealistic artifacts such as flickering between frames. By extending nuScenes with photorealistic and temporally coherent adversarial sequences, the Challenger dataset provides a robustness benchmark that remains compatible with established evaluation protocols.

In this thesis, the adversarial nuScenes dataset served as the main validation dataset to assess whether the enhancements introduced to the baseline models improved adversarial resilience without compromising general performance. All architectural modifications, robustness strategies, and optimization refinements were evaluated on the Challenger dataset to quantify their impact under adversarial driving conditions.

In addition to evaluation, the dataset was used for controlled preliminary training experiments. Before deploying any modification to our own generated dataset, we first trained and validated the proposed changes using the Challenger dataset. This two-stage protocol ensured that only empirically validated improvements were transferred to the final dataset used in this thesis. However, we intentionally avoided final-stage training on Challenger. Since it is fundamentally designed as an adversarial validation benchmark, extensive training on it could bias the models and reduce its effectiveness for validation. This strategy allowed us to

verify robustness gains during development while maintaining a reliable unseen evaluation setting for final performance assessment.

4.1.3 Challenger-Collision (CC) Dataset

Our dataset was designed as a targeted adversarial collision benchmark for autonomous driving and is directly inspired by the Challenger dataset [35]. Since Challenger itself is constructed on top of nuScenes, our dataset inherits the same general structure, data organization, and core characteristics of the nuScenes framework. This means that the dataset remains compatible with the scene-based organization, and annotation conventions commonly used in nuScenes. Preserving this structure was important because it allowed the dataset to remain consistent with established training and evaluation pipelines while introducing a more robust scenario design.

The main distinction of this dataset is that it concentrates specifically on safety-critical collision situations caused by a single adversarial vehicle interacting with the ego vehicle, it focuses on close-conflict scenarios in which the ego vehicle must react correctly under time-critical conditions. This design choice was motivated by the observation that many autonomous driving models perform well on standard benchmark scenarios yet may still fail in rare but highly consequential situations involving imminent collision risk. By explicitly constructing such cases, the dataset provides a controlled environment for evaluating whether a model is capable of safe and robust behavior when exposed to adversarial traffic participants.

To achieve this, the dataset was centered around two main scenario types: stationary and frontal collision situations. In the stationary scenario, a target vehicle is placed directly in the ego lane and remains static throughout the sequence. Although this configuration is comparatively simple in terms of target motion, it creates an important emergency response problem for the ego vehicle. Because the obstacle is already occupying the lane, the ego vehicle must detect the threat early and choose an appropriate avoidance strategy. Depending on the precise placement and orientation of the target vehicle, a safe response may consist of a harsh braking maneuver, an evasive steering maneuver, or a combination of both. This scenario is particularly valuable because it tests whether the driving model can recognize a blocked-lane hazard and execute a timely and stable avoidance action instead of reacting too late or following a nominal trajectory into collision [26].

The frontal scenario is more challenging and represents a head-on conflict situation. In this case, the adversarial target vehicle approaches from the opposite direction but has drifted into the ego lane, thereby creating a direct collision path with the ego vehicle. Unlike the stationary case, braking alone is generally insufficient to prevent a crash, since deceleration can only reduce the impact speed rather than eliminate the conflict entirely. To fully avoid collision, the ego vehicle must perform a steering-based evasive maneuver. This makes the frontal scenario especially important for evaluating whether an autonomous driving model can reason beyond longitudinal control and execute an effective lateral avoidance strategy under severe time pressure.[26].

The dataset consists of a total of 100 scenes derived from Challenger scenes. These are evenly divided between the two scenario categories, with 50 scenes corresponding to stationary scenarios and 50 scenes corresponding to frontal collision scenarios. Each scene has a duration

of five seconds and contains 11 frames. This results in an effective frame rate of approximately 2 frames per second (FPS), which closely matches the keyframe sampling frequency used in the nuScenes dataset. These two scenarios were deliberately chosen because together they capture two distinct dimensions of emergency driving behavior. The stationary scenario evaluates obstacle perception, blocked-path reasoning, and the balance between braking and steering in the presence of a static hazard. The frontal scenario, on the other hand, evaluates dynamic threat assessment, urgency-aware planning, and the ability to perform decisive lateral avoidance against an actively approaching adversarial vehicle. By limiting the dataset to a single adversarial vehicle, the interaction remains interpretable and controlled, which is beneficial for systematic analysis. At the same time, the resulting situations are sufficiently difficult to expose weaknesses in perception, prediction, planning, and end-to-end control.

The importance of such a dataset lies in the fact that conventional autonomous driving benchmarks often emphasize average-case performance across broad urban driving conditions, while safety-critical edge cases remain relatively underrepresented. However, in practical deployment, the reliability of an autonomous driving system is determined not only by how well it performs under normal conditions, but also by how safely it responds in rare, adversarial, and collision-imminent scenarios. A model that achieves strong benchmark scores may still be unsafe if it cannot handle lane blockage or head-on intrusion events.

Moreover, these scenarios are needed because they reveal whether a model has learned truly defensive driving behavior or has merely optimized for nominal benchmark statistics. In standard datasets, models may rely heavily on common traffic patterns and may not develop sufficiently strong responses to unusual hazards. In contrast, our dataset forces the model to confront situations in which immediate risk mitigation is required. This makes it particularly useful for assessing whether fine-tuning and architectural enhancements actually improve safety behavior rather than only improving aggregate performance metrics.

Overall, our dataset serves as a focused adversarial robustness benchmark that preserves the structural advantages of Challenger and nuScenes while introducing specialized collision scenarios that are highly relevant to autonomous driving safety. By emphasizing stationary lane-blocking hazards and frontal collision-path intrusions, it enables a more rigorous evaluation of emergency decision-making, evasive maneuvering, and safety-aware planning. As such, it provides an important complement to standard autonomous driving benchmarks and supports a more realistic assessment of model reliability in critical situations.

4.1.4 NeuroNCAP Evaluation Framework

In addition to the quantitative metrics, the models were further evaluated using the NeuroNCAP framework [26]. NeuroNCAP is a closed-loop evaluation framework designed to assess the safety performance of autonomous driving systems in realistic driving scenarios. Unlike traditional open-loop metrics that measure trajectory prediction accuracy in isolation, NeuroNCAP evaluates the full driving behavior of an autonomous system during simulated rollouts. This allows the evaluation process to capture how perception, prediction, planning, and control components interact when the model is responsible for making sequential driving decisions.

The NeuroNCAP framework is inspired by the New Car Assessment Program (NCAP) used in the automotive industry, where vehicles are evaluated through standardized safety tests

such as collision avoidance scenarios. Similarly, NeuroNCAP adapts this philosophy to autonomous driving models by measuring their ability to safely navigate safety-critical situations within a simulation environment. In this framework, a trained autonomous driving model receives sensor inputs generated from a photorealistic simulator and produces control commands or planned trajectories. The simulator then updates the state of the ego vehicle and surrounding actors, generating the next set of observations. This process continues iteratively, forming a closed-loop interaction between the driving model and the environment.

During evaluation, the model is exposed to predefined safety scenarios that are designed to test its ability to avoid hazardous events such as collisions with other vehicles, pedestrians, or obstacles. Because the evaluation occurs in a closed-loop setting, errors made by the model can accumulate over time, making the evaluation more representative of real-world deployment conditions compared to purely open-loop trajectory evaluation.

In the context of this thesis, the NeuroNCAP framework was used to evaluate the baseline autonomous driving models after they were fine-tuned using the proposed adversarial dataset. This allowed us to assess whether the introduced enhancements improved the models' ability to safely handle adversarial collision scenarios in a realistic closed-loop driving environment.

4.2 Evaluation Metrics

To quantitatively evaluate the performance and robustness of the autonomous driving models considered, two primary evaluation metrics were used: the L2 trajectory error and the collision rate. These metrics capture two complementary aspects of autonomous driving performance. The L2 error measures the accuracy of the predicted ego trajectory compared to the ground-truth trajectory, while the collision rate evaluates the safety of the resulting driving behavior by measuring how often the ego vehicle collides with surrounding actors.

4.2.1 L2 Trajectory Error

The L2 trajectory error measures the deviation between the predicted trajectory of the ego vehicle and the ground-truth trajectory over a given time horizon. This metric is widely used in autonomous driving and trajectory prediction tasks because it directly quantifies how closely the predicted motion follows the actual reference trajectory [39].

Let $\mathbf{p}_t = (x_t, y_t)$ denote the ground-truth position of the ego vehicle at time step t , and let $\hat{\mathbf{p}}_t = (\hat{x}_t, \hat{y}_t)$ represent the predicted position at the same time step. The L2 error is defined as the Euclidean distance between these two positions. Over a trajectory consisting of T time steps, the average L2 error can be computed as:

$$L2 = \frac{1}{T} \sum_{t=1}^T \sqrt{(x_t - \hat{x}_t)^2 + (y_t - \hat{y}_t)^2}$$

This metric captures the overall trajectory deviation across the prediction horizon. Lower L2 error values indicate that the model is producing trajectories that more closely follow the ground truth, reflecting more accurate motion prediction or planning behavior.

In the context of this thesis, the L2 trajectory error is used to evaluate how well the modified models preserve their ability to generate realistic and stable driving trajectories when exposed to adversarial scenarios. Maintaining low trajectory error ensures that improvements in robustness do not come at the cost of degraded trajectory accuracy.

4.2.2 Collision Rate

The collision rate metric is used to measure how frequently the ego vehicle collides with other actors during the evaluation scenarios. This metric is particularly relevant for adversarial datasets, where the goal is to test whether the model can successfully avoid dangerous situations.

The collision rate is defined as the proportion of evaluated scenes in which the ego vehicle experiences at least one collision event with another object or vehicle [3]. If N denotes the total number of evaluated scenes and C represents the number of scenes that contain a collision involving the ego vehicle, the collision rate can be expressed as:

$$\text{Collision Rate} = \frac{C}{N}$$

A lower collision rate indicates safer driving behavior and a stronger ability to handle adversarial or safety-critical situations. In the context of our adversarial dataset, this metric is particularly important because it directly reflects whether the autonomous driving model can successfully avoid the collision scenarios introduced by the adversarial vehicle.

Together, the L2 trajectory error and collision rate provide a balanced evaluation framework. The L2 error measures trajectory accuracy and motion consistency, while the collision rate measures safety and robustness in challenging driving conditions. Using both metrics allows us to assess whether the proposed enhancements improve the models' ability to handle adversarial scenarios without degrading their trajectory quality.

4.2.3 NCAP Score

To quantify the safety performance of the evaluated models, NeuroNCAP introduces the NCAP Score [26]. The NCAP score is a normalized safety metric that summarizes how successfully the autonomous driving system completes the evaluation scenarios without causing collisions or other safety violations. It is inspired by traditional automotive safety ratings used in regulatory vehicle testing.

The NCAP score is computed by aggregating performance across multiple evaluation scenarios and assigning penalties for unsafe outcomes. Each scenario contributes to the final score based on whether the ego vehicle successfully avoids a collision and completes the scenario safely. Scenarios in which the ego vehicle avoids collisions and maintains stable driving behavior contribute positively to the score, while collisions or unsafe maneuvers reduce the score.

Formally, the NCAP score can be expressed as a normalized ratio between the total safety points obtained during evaluation and the maximum achievable score across all scenarios:

$$NCAP\ Score = \frac{\sum_{i=1}^N S_i}{N}$$

where N represents the total number of evaluation scenarios and S_i represents the safety score obtained in scenario i . Each scenario score reflects whether the ego vehicle successfully avoided collisions and maintained safe driving behavior throughout the simulation.

The NCAP score provides an interpretable safety-oriented metric that complements trajectory accuracy metrics such as L2 error. While trajectory-based metrics evaluate how closely a model follows reference trajectories, the NCAP score focuses on the overall safety outcome of the driving behavior.

4.3 Experiments

In this section, we present the experimental evaluation of the considered end-to-end autonomous driving models and analyze their performance across multiple datasets and training configurations. The goal of these experiments is to assess how well the models generalize to safety-critical scenarios and how effectively the proposed finetuning strategies improve their behavior in adversarial driving situations.

The evaluation is conducted progressively across different datasets. We first examine the baseline performance of the pretrained models on the standard benchmark dataset in order to establish a reference for their original capabilities. We then evaluate the models on an adversarial validation dataset to analyze their robustness when exposed to collision-prone scenarios. Finally, we assess the performance of the models on the generated adversarial dataset introduced in this work.

In addition, several finetuning configurations are explored to determine which adaptation strategy best enables the models to learn the desired collision-avoidance behavior. By comparing different training setups and model variants, the experiments aim to identify the configurations that most effectively adapt the autonomous driving models to the adversarial scenarios present in the generated dataset.

4.3.1 Baseline Model Evaluation

To establish a reliable reference point for the experiments in this work, baseline evaluations were conducted for the three end-to-end autonomous driving models considered in this study: UniAD [3], SparseDrive [15], and DiffusionDrive [16]. Instead of directly reporting the performance numbers published in the original papers, we reproduced the baseline results by performing validation experiments using the official pretrained checkpoints provided by the respective authors.

The validation process was first carried out on the nuScenes validation split dataset [1] in order to verify that the reproduced results were consistent with those reported in the original works. This step ensured that the models were correctly implemented and that the experimental setup matched the expected benchmark performance. After confirming this consistency, additional validation experiments were conducted on the Challengers dataset [35]. Since this

dataset is later used as the primary validation set for evaluating the finetuned models, it was important to assess the baseline behavior of the models on it before applying any modifications or finetuning strategies.

During these baseline experiments, the original model configurations were preserved to maintain consistency with the published architectures and training setups. The only adjustments introduced were minor resource-related optimizations required to accommodate the available computational resources. These changes were limited to aspects such as batch size and training configuration parameters and did not alter the underlying model architectures or inference behavior. Any modifications made for resource management purposes are described in the following paragraphs.

SparseDrive Baseline

For the baseline evaluation of the SparseDrive model, we used the official configuration provided by the authors with minimal modifications required to accommodate the available computational resources. The evaluation was initialized from the pretrained Stage-2 checkpoints released by the original authors, which correspond to the fully trained model used in the reported experiments. Starting from these checkpoints ensures that the reproduced results reflect the intended performance of the model without requiring retraining from earlier stages.

The experiments were conducted using two GPUs with a total batch size of two samples, corresponding to one sample per GPU. During baseline evaluation, all tasks were enabled, allowing the model to jointly perform perception, prediction, and planning in a unified end-to-end framework. This configuration ensures that the reproduced baseline results reflect the full capabilities of the original SparseDrive architecture.

DiffusionDrive Baseline

For the baseline evaluation of the DiffusionDrive model, we used the official model configuration as the starting point and initialized the experiments from the pretrained Stage-2 checkpoint provided by the authors. This checkpoint corresponds to the fully trained version of the model and was used to reproduce the baseline planning performance before applying any finetuning or architectural modifications. Starting from the released checkpoint ensured that the reported results reflect the intended pretrained behavior of DiffusionDrive rather than a partially retrained variant.

To accommodate the available computational resources, the original large-scale training setup was simplified for our evaluation environment. The experiments were conducted on a single GPU with a total batch size of two samples.

UniAD Baseline Configuration

For the baseline evaluation of the UniAD model, we adopted the official stage-2 end-to-end configuration and initialized the experiments from the pretrained checkpoint released by the authors. Using this pretrained checkpoint allowed us to reproduce the model’s baseline behavior under the original unified perception, prediction, and planning framework before applying any task-specific finetuning strategies.

The experiments were conducted with a batch size of one sample per GPU and eight data-loading workers. This configuration was used to establish the baseline performance of UniAD before introducing adversarial finetuning. By preserving the original stage-2 setup and starting from the released checkpoint, the resulting baseline serves as a reliable reference for later comparisons on the Challengers and generated adversarial datasets.

4.3.2 Methodology Experiments

In this section, we describe in detail the experimental configurations and training procedures used to adapt the considered autonomous driving models to the proposed dataset. The objective of these experiments is to systematically investigate how different configuration choices influence the ability of the models to learn from the generated adversarial scenarios and to identify the most effective setup for integrating the dataset into the training pipeline.

The following subsections present the modifications introduced to the original training configurations of the models in order to accommodate the newly generated dataset within the training loop. These include adjustments related to dataset integration, training schedules, freezing strategies, and planning-related parameters. To better understand the impact of these design choices, we also conduct an ablation study that explores multiple intermediate configurations tested throughout the development process.

SparseDrive Fine-tuning

For the finetuning experiments with SparseDrive, we initialized the model from the official Stage-2 pretrained checkpoint. The goal of the finetuning process was to adapt the planning behavior of the model to the adversarial scenarios introduced in the generated dataset while preserving the pretrained perception and scene understanding capabilities.

The training setup was configured to run on two GPUs with a total batch size of two samples, resulting in one sample processed per GPU. The model was trained for 30 epochs using the AdamW optimizer with an initial learning rate of 3×10^{-4} and a weight decay of 1×10^{-3} . Gradient clipping with a maximum norm of 10 was applied to improve training stability. A cosine annealing learning rate scheduler with linear warmup over the first 500 iterations was used, gradually reducing the learning rate during training.

To focus the training process on the planning component, a selective freezing strategy was adopted. The image backbone, feature pyramid network, depth branch, detection head, and map head were frozen during finetuning by assigning a learning rate multiplier of zero to their parameters. During evaluation, only the planning module was activated while the detection, tracking, mapping, and motion prediction outputs were disabled. This configuration allows the experiments to directly assess the impact of the finetuning process on the trajectory planning behavior of the model under adversarial driving scenarios.

DiffusionDrive Finetuning Configuration

For the finetuning experiments with DiffusionDrive, the model was initialized from the official Stage-2 pretrained checkpoint. The objective of this setup was to adapt the diffusion-based planning behavior of the model to the generated adversarial scenarios while preserving

the pretrained perception, mapping, and interaction reasoning components learned from the original dataset.

The training was conducted on two GPUs with a total batch size of two samples, corresponding to one sample per GPU, and was run for 30 epochs. Optimization was performed using the AdamW optimizer with a learning rate of 3×10^{-6} and a weight decay of 1×10^{-3} . A cosine annealing learning rate schedule with linear warmup over the first 500 iterations was used, and gradient clipping with a maximum norm of 10 was applied to improve training stability. Compared with SparseDrive, a substantially smaller learning rate was adopted in order to preserve the stability of the diffusion-based planning process during finetuning.

As in the SparseDrive setup, the finetuning strategy was designed to focus on the planning task only. The image backbone, image neck, depth branch, detection head, and map head were frozen by assigning a learning rate multiplier of zero to their parameters. This evaluation protocol isolates the planning performance of DiffusionDrive and allows the effect of the finetuning strategy on collision-avoidance trajectory generation to be assessed directly.

UniAD Finetuning Configuration

For the finetuning experiments with UniAD, the model was initialized from the pretrained end-to-end checkpoint. The aim of this setup was to adapt the planning behavior of UniAD to the generated adversarial scenarios while preserving its pretrained perception, tracking, mapping, occupancy, and motion reasoning capabilities. Since UniAD follows a unified multi-task architecture, the finetuning strategy was designed to restrict learning primarily to the planning component so that the model could adjust its ego trajectory generation without altering the rest of the pipeline.

Training was performed for 5 epochs with a batch size of one sample per GPU. The model was optimized using AdamW with a learning rate of 1×10^{-5} and a weight decay of 1×10^{-3} . A cosine annealing learning rate scheduler with linear warmup over the first 500 iterations was used, and gradient clipping with a maximum norm of 2 was applied to stabilize optimization. The use of a reduced learning rate was intended to preserve the stability of the pretrained UniAD representation while enabling the planning module to adapt to the newly introduced adversarial driving situations.

A selective freezing strategy was again adopted. The image backbone, image neck, batch normalization layers, and BEV encoder were kept frozen, consistent with the pretrained UniAD setup.

Training Dataset Composition

All methodology experiments in this work were conducted using a mixed training dataset composed of the generated adversarial dataset and the original nuScenes dataset. In the final setup, 15% of the training data was taken from the generated dataset, while the remaining 85% was sampled from nuScenes. This ratio was adopted consistently across all finetuning experiments, as it provided the most effective balance between adversarial adaptation and preservation of general driving behavior, a result that will later be supported by the ablation study.

Using only the generated dataset for finetuning was found to be insufficient because of its relatively small size and its exclusive focus on adversarial scenes. Training solely on these scenarios caused the models to over-specialize to collision-prone interactions and reduced their ability to maintain appropriate behavior in normal driving situations. For example, the models became less consistent in preserving lane-following behavior when no immediate obstacle was present and were less exposed to the variability of ordinary traffic patterns. Incorporating a larger proportion of nuScenes data therefore helped preserve standard driving priors, while the adversarial subset encouraged the models to learn stronger collision-avoidance responses. As a result, the mixed-dataset strategy provided a more suitable training distribution for all methodology experiments.

For validation, the Challengers dataset was used as the primary validation set. This dataset represents an intermediate level of difficulty between the normal traffic scenarios present in nuScenes and the highly adversarial scenes introduced in the generated dataset. In particular, Challengers contains close-collision situations that require corrective maneuvers, but the scenarios remain relatively moderate and avoidable compared to the intentionally harsher adversarial cases in our dataset. Therefore, it serves as a suitable validation benchmark that evaluates the model’s ability to handle safety-critical situations while still reflecting realistic driving behavior.

5 Results

This chapter presents the results obtained from the experiments described in the previous sections. The performance of the evaluated end-to-end autonomous driving models is analyzed using the quantitative metrics introduced earlier. In addition to reporting numerical results, we provide a qualitative analysis through visual examples of predicted trajectories and BEV representations. These visualizations help illustrate how the models react to the introduced adversarial scenarios and provide further insight into their behavior beyond the reported metrics.

5.1 Baseline Results

The baseline results presented in Table 5.1 are intended to provide a reference point for the subsequent experiments. The primary objective of these baseline evaluations is to identify the relative performance of the models before applying any finetuning or dataset augmentation strategies. In particular, this allows us to determine which model exhibits the highest vulnerability to collision scenarios and therefore provides the most suitable starting point for improvement.

From the reported results, the UniAD model demonstrates the highest collision rates across both evaluated datasets. This observation is consistent with the performance trends reported in the original UniAD paper and can also be attributed to the fact that UniAD represents one of the earlier end-to-end autonomous driving architectures among the evaluated models. In contrast, the best performing model on the nuScenes dataset is DiffusionDrive, which achieves the lowest collision rate under normal traffic conditions. For the Challengers adversarial dataset, SparseDrive reports the lowest collision rates, indicating a stronger robustness when exposed to close-collision scenarios. These differences can be explained by the architectural evolution of the models, as more recent approaches such as SparseDrive and DiffusionDrive were developed as improvements over earlier frameworks like UniAD, incorporating refinements in perception, motion prediction, and planning modules.

These baseline observations therefore provide a useful starting point for the subsequent finetuning and adversarial robustness experiments presented in the following sections.

5.1.1 Methodology Results

Table 5.2 reports the collision rate of the finetuned models evaluated on the Challengers dataset. To better quantify the improvement obtained through finetuning, the percentage decrease in the average collision rate relative to the baseline results is also reported.

Dataset	AD Model	1s	2s	3s	avg	AD Model	1s	2s	3s	avg
NuScenes	UniAD [3]	0.13%	0.18%	0.66%	0.26%	SparseDrive [15]	0.01%	0.05%	0.23%	0.09%
Challengers (AdvNuSc)		2.76%	7.36%	9.80%	5.97%		0.01%	0.62%	2.43%	1.03%
NuScenes	DiffusionDrive [16]	0.02%	0.04%	0.14%	0.07%					
Challengers (AdvNuSc)		0.07%	1.28%	3.64%	1.66%					

Table 5.1: Collision rate comparison across datasets for baseline end-to-end autonomous driving models.

AD Model	1 sec	2 sec	3 sec	Avg Col.Rate	↓ Decrease (%)
SparseDrive (Finetuned)	0.04	0.37	1.80	0.74	28.2%
DiffusionDrive (Finetuned)	0.13	0.71	1.83	0.89	46.4%
UniAD (Finetuned)	0.46	1.07	2.20	1.24	79.2%

Table 5.2: Collision rate results of finetuned autonomous driving models evaluated on the Challengers dataset. The final column reports the percentage decrease in average collision rate compared to the baseline results.

Overall, all evaluated models show a clear reduction in collision rate after finetuning on the mixed dataset. SparseDrive achieves a decrease of 28.2% in the average collision rate, while DiffusionDrive shows a larger improvement of 46.4%. The most significant improvement is observed for the UniAD model, which achieves a reduction of 79.2% in the average collision rate. This result indicates that UniAD benefits the most from the finetuning process, likely due to its higher initial vulnerability to adversarial collision scenarios in the baseline evaluation.

These results shows that the proposed dataset and finetuning strategy effectively improve the robustness of end-to-end autonomous driving models when exposed to adversarial traffic interactions.

5.2 Discussion

To further analyze the effect of the finetuning process beyond the quantitative results, a qualitative evaluation was conducted using a subset of the generated dataset that was kept aside and not used during training. This subset contains adversarial scenarios designed to trigger potential collision situations, allowing us to visually inspect how the finetuned models react when encountering an adversarial vehicle on the road. For this purpose, three representative examples shown in Figure 5.1 , 5.2 , 5.3, each corresponding to one of the evaluated models.

The visualizations illustrate the predicted ego trajectories and the corresponding BEV representations when the ego vehicle encounters an adversarial vehicle positioned in a collision-prone configuration. After finetuning, all models demonstrate an improved ability to adjust their trajectories in order to avoid the obstacle. Among the evaluated models, UniAD exhibits the most flexible avoidance maneuver. In particular, the ego vehicle first performs a slight rightward adjustment to avoid the adversarial vehicle and then smoothly returns to its original lane to continue driving straight. This behavior indicates a more adaptable planning response

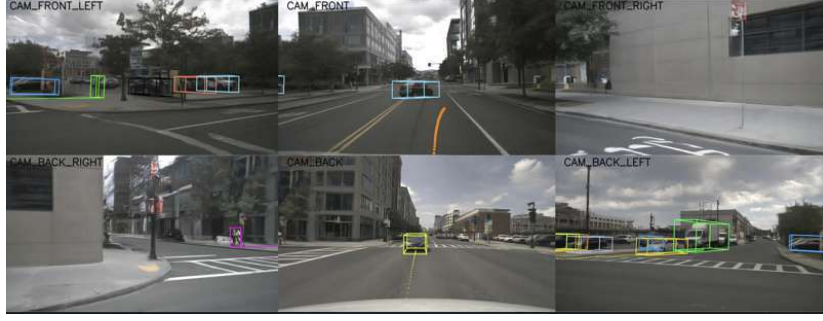


Figure 5.1: Qualitative example of the finetuned SparseDrive model in an adversarial collision scenario. The ego vehicle adjusts its trajectory to avoid the adversarial vehicle; however, the maneuver appears less flexible compared to UniAD due to the reliance on predefined driving commands that constrain the trajectory planning behavior.



Figure 5.2: Qualitative example of the finetuned DiffusionDrive model encountering an adversarial vehicle. The model attempts to adjust its trajectory to avoid the obstacle, but the resulting maneuver remains constrained by the command-based planning structure, leading to a less adaptive avoidance pattern compared to the behavior observed in the UniAD model.



Figure 5.3: Qualitative example of the finetuned UniAD model reacting to an adversarial vehicle. The predicted ego trajectory demonstrates a flexible avoidance maneuver, where the ego vehicle performs a slight rightward deviation to avoid the obstacle before smoothly returning to its original lane. This behavior highlights the model's ability to generate continuous and adaptable trajectory adjustments when responding to potential collision scenarios.

when reacting to unexpected obstacles. In contrast, SparseDrive and DiffusionDrive show less flexible avoidance patterns. While both models attempt to adjust their trajectories to prevent collision, their responses appear more constrained due to their reliance on predefined high-level driving commands, such as turning or going straight. As a result, their maneuvers tend to be less smooth and less adaptable compared to the trajectory adjustments observed in the UniAD model. These qualitative observations provide additional insight into how different architectural design choices influence the ability of end-to-end autonomous driving models to perform nuanced collision avoidance behaviors.

6 Conclusion and Future work

In this research, we proposed a novel methodology for improving the robustness of end-to-end autonomous driving models under adversarial and collision-oriented scenarios. The proposed framework focuses on generating a specialized dataset designed to expose weaknesses in existing autonomous driving systems and to guide model adaptation through targeted finetuning. The dataset was inspired by the structure of the nuScenes and Challengers datasets, while introducing adversarial interactions such as frontal and stationary near-collision scenarios that are underrepresented in standard driving datasets.

Using this dataset, we finetuned several state-of-the-art end-to-end autonomous driving models, including UniAD, SparseDrive, and DiffusionDrive. The models were trained using a mixed dataset strategy combining the generated adversarial scenes with the original nuScenes dataset, ensuring that the models preserved their ability to operate under normal traffic conditions while improving their behavior in safety-critical situations. Experimental results demonstrated that finetuning on the proposed dataset significantly reduced collision rates across all evaluated models. In particular, the UniAD model exhibited the largest improvement, achieving a substantial reduction in collision rate when evaluated on the Challengers dataset. These results highlight the effectiveness of the proposed dataset and training strategy for improving the robustness of end-to-end driving models when exposed to adversarial driving scenarios.

Furthermore, the models were evaluated within the NeuroNCAP framework, allowing their safety performance to be assessed in a standardized evaluation environment. The results indicate that targeted adversarial dataset augmentation can improve model behavior in challenging driving situations without negatively affecting general driving performance. Overall, the findings of this work demonstrate the importance of carefully designed adversarial datasets for advancing the safety and reliability of end-to-end autonomous driving systems.

Despite the encouraging results, several directions for future work remain open. First, the current dataset focuses on a limited set of adversarial collision scenarios, primarily involving frontal and stationary interactions. Future work could expand the dataset to include a broader range of safety-critical situations, such as complex multi-agent interactions, dynamic obstacle avoidance, or occlusion-based hazards. Second, the dataset generation process could be further automated using generative models or scenario generation frameworks to produce a wider variety of realistic driving scenes.

Another promising direction is the integration of more advanced video generation techniques, such as large-scale driving world models, to produce fully controllable synthetic driving environments. This would allow researchers to systematically design rare or dangerous scenarios that are difficult to capture in real-world datasets. Additionally, future work could investigate the impact of adversarial dataset generation on other components of autonomous driving systems, including perception, motion prediction, and decision-making modules.

Finally, while this work focuses on improving robustness through dataset augmentation and finetuning, further research could explore the combination of this approach with other safety mechanisms, such as uncertainty estimation, rule-based constraints, or formal verification methods. Such approaches may provide complementary strategies for ensuring safe behavior in real-world autonomous driving deployments.

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Declaration

Herewith, I declare that I have developed and written the enclosed thesis entirely by myself and that I have not used sources or means except those declared.

This thesis has not been submitted to any other authority to achieve an academic grading and has not been published elsewhere.

Stuttgart,

Samar Fathalah